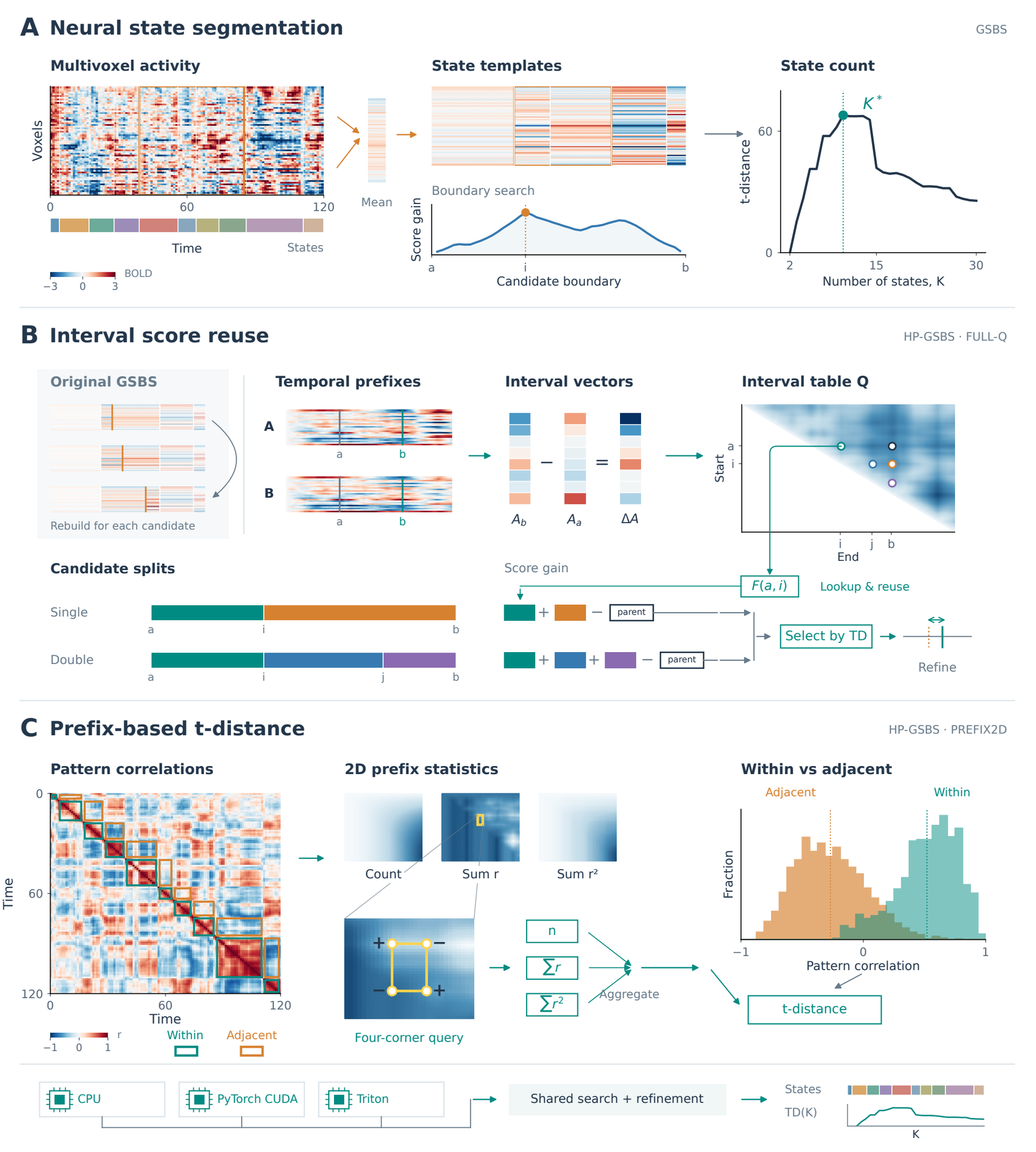

A Neural state segmentation
GSBS
Multivoxel activity
Voxels
0
60
120
Time
States
BOLD
−3
0
3
Mean
State templates
Boundary search
Score gain
a
i
b
Candidate boundary
State count
K*
60
0
t-distance
2
15
30
Number of states, K
B Interval score reuse
HP-GSBS · FULL-Q
Original GSBS
Rebuild for each candidate
Temporal prefixes
A
B
a
b
Interval vectors
A_b
A_a
ΔA
Interval table Q
Start
End
Candidate splits
Single
Double
j
Score gain
$F(a, i)$
Lookup & reuse
parent
Select by TD
Refine
C Prefix-based t-distance
HP-GSBS · PREFIX2D
Pattern correlations
Time
r
Within
Adjacent
2D prefix statistics
Count
Sum r
Sum r²
Four-corner query
n
$\sum r$
$\sum r^2$
Aggregate
Within vs adjacent
Fraction
Pattern correlation
t-distance
CPU
PyTorch CUDA
Triton
Shared search + refinement
States
TD(K)
K